Popular Translation Textual Analysis

Team Members: Sarah Brown and Brandon Duong

Table of Contents

STEM INTRODUCTION 1
TEXT INTRODUCTION 1
TEXT ANALYSIS 1
Ocean's 8 1
Example 1 2
Example 2 2
Westworld 4
Example 1 4
Example 2 4
Lost in Space 5
Example 1 5
Example 2 6
SOCIAL MEDIA ANALYSIS 8
Long Form Example: YouTube 8
Short Form Example: TikTok/Instagram Reels 9
Best Practices 9
CONCLUSION 10

STEM INTRODUCTION

3D printing is the process of making a physical object from a three-dimensional digital design. This process is usually done by 'printing' thin layers of material over each other and fusing them. When most people think about 3D printing, they probably think about making little toys and figures to mess around with or display but its use has much more potential than that. 3D printing can be and is used in multiple different industries and applications. From healthcare, fashion, design, and even personal use, the versatility of 3D printing is immense. While 3D printers have existed for just about 4 decades now, they have existed in fiction for much longer (Adlughmin). One of the earliest mentions of creating objects via machines like 3D printers was 83 years ago in a novelette titled "The 4-Sided Triangle" by William F. Temple ("What Is the First Reference to a 3D Printer in Science Fiction?"). However, as fiction has turned to fact, it is important to make note of what is actually possible and what still remains outside of our, current, technological grasp.

TEXT INTRODUCTION

While 3D printers have existed for just about 4 decades now, they have existed in fiction for much longer (Adlughmin). However, 3D printers have existed in fiction for much longer. One of the earliest mentions of creating objects via machines like 3D printers was 83 years ago in a novelette titled "The 4-Sided Triangle" by William F. Temple ("What Is the First Reference to a 3D Printer in Science Fiction?"). However, as fiction has turned to fact, it is important to make note of what is actually possible and what still remains outside of our, current, technological grasp. We will analyze the following texts, *Ocean's 8*, *Westworld*, and *Lost in Space*. As a side effect, spoilers for these texts may exist. A lot of media currently is a mix of fact and fiction; the bridge between what we know is possible and what we cannot quite reach yet. One example of this is *Ocean's 8.* In *Ocean's 8*, the heist crew scans and then 3D prints a valuable necklace so as to swap it with the real necklace during a banquet scene. On the other hand, some current science-fiction shows are currently very far from feasible such as *Westworld* where bionic robot flesh is printed in mass and with rapid speed. However, while the rapid printing and scale that is seen in this show is not currently available, 3D printed muscles and organs could be a much more feasible goal. Finally, *Lost in Space* shows uses of 3D printing that are not only useful, but are currently possible. Tools are printed to help characters as they travel through space. While the space travel aspect of *Lost in Space* is not currently achievable, 3D printing has been tested in space on the International Space Station and is possible within our current grasp of the technology.

TEXT ANALYSIS

Ocean's 8

3D printing features strongly as a plot point in *Ocean's 8.* The characters in this movie attempt to carry out a heist to steal a priceless necklace during the Met Gala. To avoid detection, they 3D print a copy. The necklace is replicated by first being 3D scanned and then the created model is 3D printed. These two examples will be explored further below.

Example 1

The first example is centered on a scene where the characters of the movie use the material zirconium to 3D print a replica diamond necklace (Jackson). This replica necklace is then used as a decoy to delay detection and thus give the heist team time to distance themselves from the scene. While 3D printing technology to replicate diamonds is in the works, the consensus is that it isn't quite possible yet. Specifically, the replicating the shine of diamonds would be close to impossible due to the layers created as the object is printed (Jackson). However, that didn't stop Warner Bros from trying to accurately represent the technology by working with MakerBot, a 3D printer company. "When Warner Bros approached us with the pitch for Ocean's 8, we were thrilled to introduce them to the Replicator Z18 and give our best tips for how to shoot the printer without adding too much CG. That's always a big concern — especially with the history of hype and media portrayals creating unrealistic expectations of the technology" (Goehrke). While this example had some incorrect aspects, it overall accurately represented the growth and possibilities of 3D printing as a whole. Unfortunately, viewers won't be able to complete elaborate heists anytime soon with the help of 3D printers, but *Ocean's 8* works as a good introduction to creative uses of the technology.



Figure 1: The 3D printer used is shown to the left while Sandra Bullock's character Debbie Ocean holds up one of the 3D printed diamonds used to create the replacement necklace.

Example 2

Before the necklace could be printed, it was first scanned. 3D Scanning is a process used to collect information about our surroundings and our world. The collected information can be used to replicate things as 3D model. This process can be done in various different ways including time-of-flight, laser scanners, and structured lighting. This method is closely tied to 3D printing as the created models can then be printed. "Florian Bommier, application engineer at SHINING 3D, suggests a plan for the design of the glasses; 'cameras would be on the side of the glasses,' i.e. hidden in the classic pins either side of Wayfarer glasses, 'and the projector on the center'" (Jackson).



Figure 2: The glasses used to 3D scan the diamond necklace with built-in cameras in the frames of the glasses.

However, the shine of the diamonds poses an issue as it could disrupt a scan of the necklace using lasers or infrared light sources. As such, photogrammetry is a suggested alternative. “CAD sculptor and jewelry designer Morgan Morey, winner of the 2017 3D Printing Industry Awards trophy design competition, also backs the idea. Making a CAD model of a diamond necklace according to Morey, ‘takes a matter of hours and would only need a half decent picture or diagram’” (Jackson). Therefore, the 3D scanning of the necklace would be very feasible. Although, it would be a lot easier in a lower stakes situation than planning a heist, such as for historical preservation. 3D scanning and 3D printing can be used in places like museums to replicate objects that visitors can touch. 3D scanning is not as polished as shown in *Ocean’s 8,* but this scene serves as a good introduction to existing technology.



Figure 3: 3D scanning process of the diamond necklace scanned by Helena bonham Carter’s character Rose Weil.

Westworld

In *Westworld* 3D printing is a core technology used to make a Western resort with A.I. hosts possible. These A.I. hosts are made of 3D printed biological materials and the plot resolves around their quest for sentience and the moral dilemmas that arise as a result. As 3D printing is more of a background technology than the main focus of the show, there will be some overlap between these two examples. Example 1 will focus on the 3D printing featuring in the show's opening credits while example 2 will focus on one of the specific scenes where 3D printing is used.

Example 1

Westworld'*s* opening credits heavily feature the use of 3D printing set to a thoughtful piano piece. These scenes are used to set the scene of the TV show in a Western setting while also reinforcing the fact that nothing is as it seems. A.I. hosts, animals, and background props are all created via 3D printing and are available for the human visitors to Westworld to use at their discretion. The technology in *Westworld* is still very far from feasible as we are nowhere near being able to 3D print bionic robot flesh in mass or with rapid speed. However, while the rapid printing and scale that is seen in this show is not currently available, 3D printed muscles and organs could be much closer. Back in 2016, MIT developed a type of plastic muscle that researchers were able to control via heat (Liszewski). In addition, researchers at Columbia have also been making strides in research about artificial muscles (Evarts). While the technology doesn't exist yet, innovations in the research field are constantly improving. Overall, Westworld focuses more on the ethical questions posed by A.I. and the 3D printing aspect is just something that allows the A.I. to exist in our world.



Figure 4: A screenshot from Westworld's opening credits showing the 3D printing of a horse.

Example 2

While neither the current A.I. nor the 3D printing technology is at the level shown in the show, it poses important questions to viewers about ethics and things that need to be considered as research continues. "The opening sequence visually describes the construction of the robotic hosts. Instead of a mechanical assembly process, the hosts are seen being created using futuristic

3D printing and other additive manufacturing techniques. And it's not just plastic or metals as the base material but organic material as well. Even today researchers are exploring 3D printing individual body parts. What would stop us from eventually putting all these parts together to create a complete life, human or animal?" (Parks). Not only are viewers challenged by questions of whether or not this technology is possible, but *Westworld* introduces the question of *should* this technology be possible.

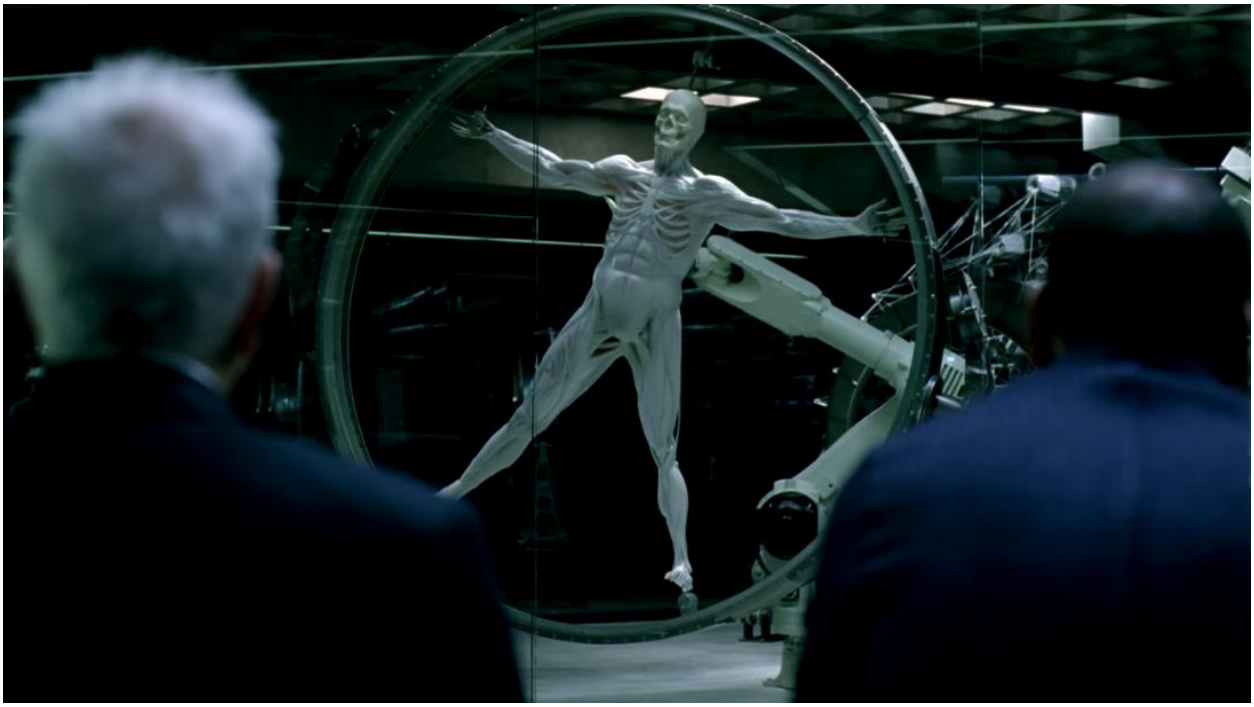


Figure 5: The body of an A.I. host is printed while workers of the park Westworld watch and discuss.

Lost in Space

In *Lost in Space* 3D printing is assumed to be a common tool, there to be used and is a part of the toolbox that the characters use throughout their space travels. Two examples of this that will be explored further are when a medical brace is printed for a character's arm and a casual reference to printing replacement parts.

Example 1

Tools and medical devices (such as a leg braces) are printed to help characters as they travel through space. This representation is not only correct, but also is something that current astronauts are using on the International Space Station where a 3D printer has been installed (Gaskill). This is a valuable depiction of 3D printing as it not only shows how 3D printing can be used on Earth today, but also shows how it could impact our future in space. "Despite taking place on an alien planet in 2048, the Robinson's 3D printer seems to be afflicted by many of the same limitations as today's machines … A particularly impressive detail is shown early on, when a leg brace is printed for an injured crew member. Due to the shape of the brace, a substantial amount of support material was necessary. The support is a different color than the printed piece, and sure enough, the printer is shown with dual extruders. Even the positioning and shape of the support columns properly corresponds with the overhang on the object" (Nardi).



Figure 6: A 3D printed leg brace used to support an injured character.

Example 2

In addition to accurately representing current 3D technology, *Lost in Space* attempts to show that it is normalized in the show's society. When the characters in *Lost in Space* need tools or any sort of spare parts they simply select a model from the ship's library and set it up to print. "Characters simply remark that they are going to go print some parts, or else ask another character to print something for them. No one ever asks for clarification or needs an explanation on how to do it; it's a technology so ingrained in their life that even the children in the show are familiar with it" (Nardi). Overall, *Lost in Space*'s representation of 3D printing is accurate and also helps normalize its use in both the show's setting, but also helps to normalize it in our world today. Viewers are exposed to 3D printing in a way that is both helpful and every day.

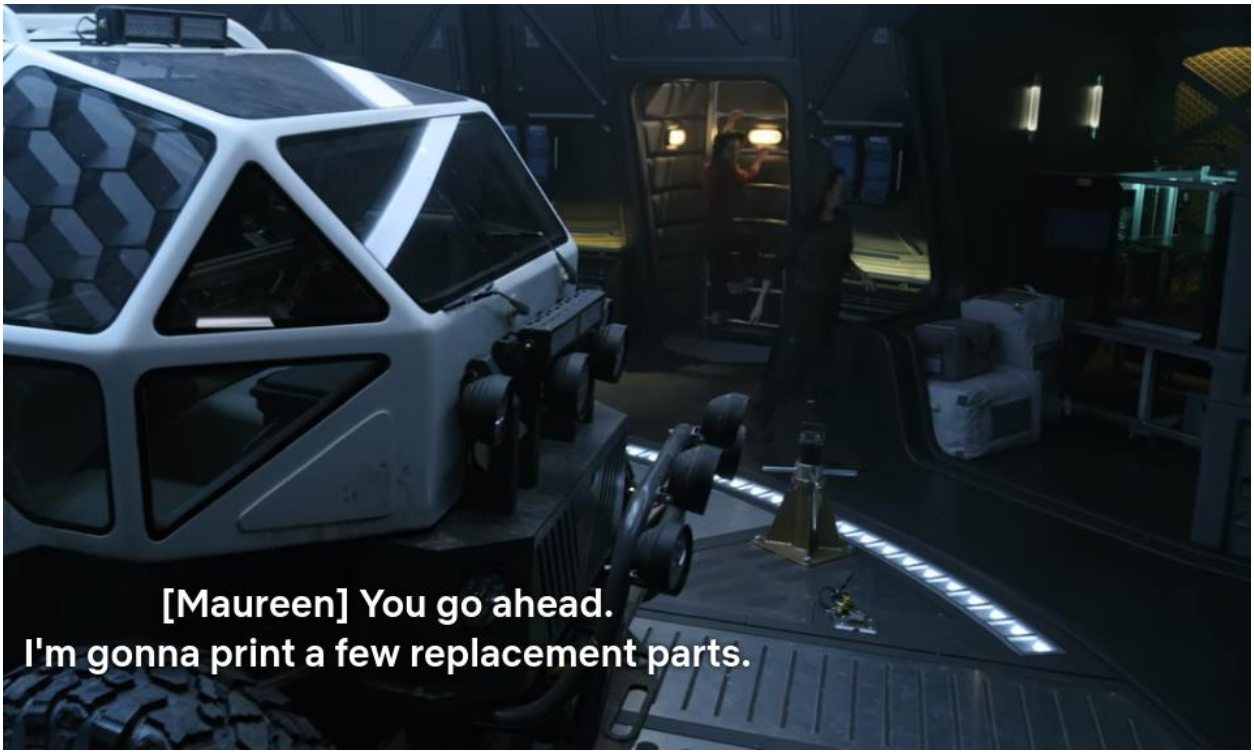


Figure 7: An offhand remark to 3D printing in the show Lost in Space.

SOCIAL MEDIA ANALYSIS

Long Form Example: YouTube

With a complex topic such as 3D Printing, YouTube is a great way to create amazing informational content that almost anyone can have access to. One of the strengths of YouTube is the ability to pack large amounts of information into an intriguing and captivating form factor that is able to grab anyone's attention. I am sure everyone reading this document has gone on YouTube before to find a recipe video or DIY guide and found themselves suddenly looking at puppy videos. That is the power of YouTube and something that we would hope to capitalize on.



Figure 8: Screenshot from a YouTube Video called "Is 3D printing a revolution or just a trend?" by Interesting Engineering. The video talks about the history of 3D printing in both mainstream media and real-world applications.

In the YouTube video "Is 3D printing a revolution or just a trend?", the video talks about the present and future potential of 3D printing and how it is a practice that will only grow in the coming years. It is stated that "The Global 3D printing market size is expected to reach around $41 billion USD by 2026" (Interesting Engineering). As society changes with technology, it is important that leaders in the 3D printing industry also adapt the ways of reaching their audience. Many companies, like Prusa and Ultimaker, are now talking advantage of the YouTube space putting out great informational videos and podcasts where they speak about the 3D printing industry and what lies on the horizon (Stevenson). Coming off the pandemic, podcasts have become increasingly popular in today's society. Filming and posting the video is often the easiest way to provide that content for their audience. Videos like the one above give just enough information for the viewer to feel satisfied about their experience but also leaves enough to be desired in case they want to find out more about the topic. This tactic is seen by ending the video in rhetorical questions for the viewer to answer for themselves or create a need to search for more content on that specific YouTube channel.

Short Form Example: TikTok/Instagram Reels

Like the emergence of Podcasts and YouTube, short form content like TikTok videos and Instagram reels have seen a huge skyrocket in popularity, YouTube even has their own short form content in YouTube Shorts. Short form content allows for the creator to put out quick, interesting clips about a certain topic. Often times, the best way to inform someone about a topic is to show them real physical examples.

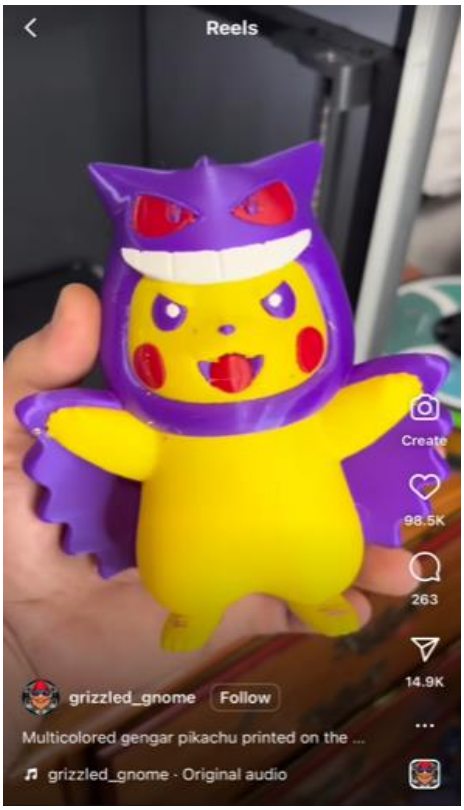


Figure 9: Instagram reel by Grizzled_Gnome who created a Pokemon Pikachu wearing a



Figure 10: TikTok made by Socal3D showing viewers 5 useful 3D prints that anyone could use.

Both figures above show two examples of short form content made on Instagram(left) and TikTok(right). These types of videos are short, fun, and grab the attention of the viewer as they are swiping though hundreds of videos. Even if one particular user is not interested in 3D printing, it is only natural that they become interested in videos like these that appeal to everyone and are not catered to people who are specifically looking for content that involves 3D printing. As we stated earlier, it is important for companies and leaders in the 3D printing industry to adapt the way they choose to attract their existing audience and potential new ones(Stevenson). These types of videos get huge engagement on their respective platforms because it approaches the topic of 3D printing in a way that is accessible to all people, which is the goal.

Best Practices

Succeeding on any social media platform is not always a simple 5 step plan, but there are always things that we can do in order to maximize the effectiveness of our posts and each platform. On

any social media platform content is king. Creating well thought out, engaging content is by far the most important step in creating a successful social media presence. For a YouTube channel, many creators follow the 80/20 rule – where 80% of social media posts either educate, entertain or help followers solve a problem and the 20% is used to promote the channel, business, website, etc. People want to feel like they are getting their content and information from a source that truly cares about the content that is it making on the platform. High quality posts are the best way to show expertise and passion for something that you want to inform people about(Wesley-Powell). This still applies to short form content but there are minor differences in the way to approach it. For example, on short form content, you might seen lots of hashtags pinned in the caption or comments. These hashtags are important and easy way to boost engagement on videos. Anyone searching for one of those tags automatically puts any video with the same tags on their search list which gives them the chance to potentially view and interact with it.

CONCLUSION

3D printing is a popular and emerging topic in both social media and industry. In most cases, social media has done a great job in depicting what 3D printing is and the history behind it. With short form content like reels, viewers can see real world ordinary applications of 3D printing which is a great way to inform and captivate new people who are unfamiliar with what 3D printing might be. In other forms of media, it is also important to show the fantasy aspect of the industry as well. People need to believe that there is a future for 3D printing beyond what we can currently produce right now. That is the only way innovation ever moves forward. In our next assignment, we'd like to potentially make a YouTube video or Instagram posts that follow these practices.

Works Cited

Adlughmin. “You Can Now See the First Ever 3D Printer - Invented by Chuck Hull - in the National Inventors Hall of Fame.” 3DPrint.Com | The Voice of 3D Printing / Additive Manufacturing, 16 Oct. 2021, https://3dprint.com/72171/first-3d-printer-chuck-hull/.

Armstrong, Katie. “Art Meets Reality in HBO's Westworld 3D Printed Inspiration.” 3D Printing Industry, 14 Oct. 2016, https://3dprintingindustry.com/news/art-meets-reality-hbos-westworld-3d-printed-inspiration-96884/.

Brayson, Johnny. “Is Crystal Printing from 'Ocean's 8' Real? Making Fake Jewelry Isn't as Far-Fetched as It Seems.” Bustle, Bustle, 8 June 2018, https://www.bustle.com/p/is-crystal-printing-from-oceans-8-real-making-fake-jewelry-isnt-as-far-fetched-as-it-seems-9288754.

Evarts, Holly. “One Step Closer to Lifelike Robots.” Columbia Engineering, 4 Mar. 2022, https://www.engineering.columbia.edu/news/one-step-closer-lifelike-robots.

Gaskill, Melissa. “Solving the Challenges of Long Duration Space Flight with 3D Printing.” NASA, NASA, 16 Dec. 2019, https://www.nasa.gov/mission_pages/station/research/news/3d-printing-in-space-long-duration-spaceflight-applications.

Goehrke, Sarah Anderson. “Ocean’s 8 and Lingering Hollywood 3D Printing Hype.” 3DPrint.Com | The Voice of 3D Printing / Additive Manufacturing, 17 Oct. 2021, https://3dprint.com/216335/oceans-8-3d-printing/.

Hilkene, Cullen. “3D Printing in Westworld: Science Fiction or Science Fact?” LinkedIn, 8 Mar. 2018, https://www.linkedin.com/pulse/3d-printing-westworld-science-fiction-fact-cullen-hilkene.

Jackson, Beau. “3D Printing and 3D Scanning Experts Crack the Ocean's 8 Diamond Necklace Heist.” 3D Printing Industry, 20 June 2018, https://3dprintingindustry.com/news/3d-printing-3d-scanning-experts-crack-oceans-8-diamond-necklace-heist-134863/.

Liszewski, Andrew. “MIT's New Plastic Muscles Could Bring Us One Step Closer to a Real-Life Westworld.” Gizmodo, Gizmodo, 23 Nov. 2016, https://gizmodo.com/mits-new-plastic-muscles-could-bring-us-one-step-closer-1789302210.

McGlade, Conor. “Is Westworld the 3D Printing of Tomorrow?” Thomasnet, https://www.thomasnet.com/insights/is-westworld-the-3d-printing-of-tomorrow-/.

Nardi, Tom. “Lost in Space Gets 3D Printing Right.” Hackaday, 25 June 2018, https://hackaday.com/2018/06/25/lost-in-space-gets-3d-printing-right/.

Parks, Mike. “Westworld: Where Technology and Ethics Collide.” Bench Talk for Design Engineers, 21 July 2017, https://www.mouser.com/blog/westworld-where-technology-and-ethics-collide.

“r/3dprinting - 3D Printing in the New Lost in Space Show.” Reddit, https://www.reddit.com/r/3Dprinting/comments/8cv2vw/3d_printing_in_the_new_lost_in_space_show/.

“r/3dprinting - I Recently Re-Watched Westworld. I Now Look at the 3D.” Reddit, https://www.reddit.com/r/3Dprinting/comments/wnnekl/i_recently_rewatched_westworld_i_now_look_at_the/.

“r/3dprinting - Lost in Space 3D Printed Buildings.” Reddit, https://www.reddit.com/r/3Dprinting/comments/rb3psh/lost_in_space_3d_printed_buildings/.

Rausch, Alex. “3D Printing in Movies: Fact or Fiction.” The 3D Printing Store, 25 May 2021, https://the3dprintingstore.com/3d-printing-in-movies-fact-or-fiction/.

“Rotten Tomatoes Lost in Space.” Rotten Tomatoes, https://www.rottentomatoes.com/tv/lost_in_space.

“Rotten Tomatoes Ocean's 8.” Rotten Tomatoes, https://www.rottentomatoes.com/m/oceans_8.

“Rotten Tomatoes Westworld” Rotten Tomatoes, https://www.rottentomatoes.com/tv/westworld.

Scott, Clare. “Westworld's Technology Is a Bit Too Familiar for Comfort: What Happens When Robotics and 3D Printing Advance Too Far?” 3DPrint.Com, 16 Oct. 2021, https://3dprint.com/151246/westworld-3d-printing-robotics/.

Sloat, Sarah. “'Lost in Space': 3D-Printed Gun Plot Isn't Futuristic, It's Already Here.” Inverse, Inverse, 19 Apr. 2018, https://www.inverse.com/article/43896-lost-in-space-3d-printed-gun.

Stevenson, Kerry. “3D Print Companies Leveraging Social Media in New Ways ", *Fabbaloo*, 25 Sept. 2021, https://www.fabbaloo.com/news/3d-print-companies-leveraging-social-media-in-new-ways.

Wesley-Powell, Ariana. “How to Boost Engagement on Social Media.” *Hurrdat Marketing*, Hurrdat Marketing, 21 June 2022, https://hurrdatmarketing.com/digital-marketing-news/ways-to-increase-social-media-engagement/.

“What Is the First Reference to a 3D Printer in Science Fiction?” Science Fiction & Fantasy Stack Exchange, 21 June 2015, https://scifi.stackexchange.com/questions/93229/what-is-the-first-reference-to-a-3d-printer-in-science-fiction.